

Flashing an OS Decoder with firmware

Prelude

Programming any OS decoder has to be done via a method which is called In Circuit Serial Programming or ICSP for short. The Arduino website has a [very elaborate explanation](#) how that works and how you can do this. I have a shorter tutorial however. It does involve using a computer.

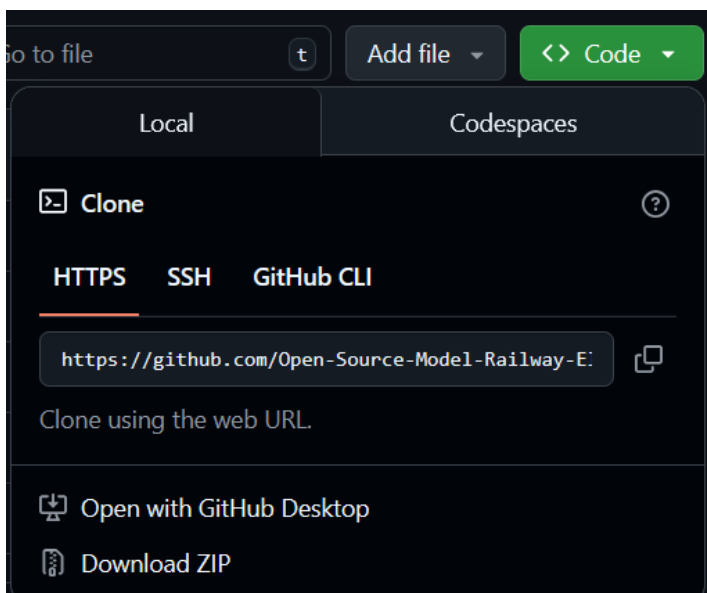
One final note. This page was originally written for the first decoder named OSSD, but the process is the same for all OS decoders and every other Open-Source design that require firmware.

Step 1: Download the necessary files.

The firmware that is going to be flashed on the OS-decoder is a so called binary file. You can recognize them by the .hex extension. All officially released binaries are to be found on this website.

<https://github.com/Open-Source-Model-Railway-Electronics/OS-software-tool>

In order to obtain the software you need to download the entire repository. You can click on the green code button and click "Download Zip"



Make the connections

An OS-decoder cannot be connected to a computer directly. In order to do this we need to use an Arduino board.

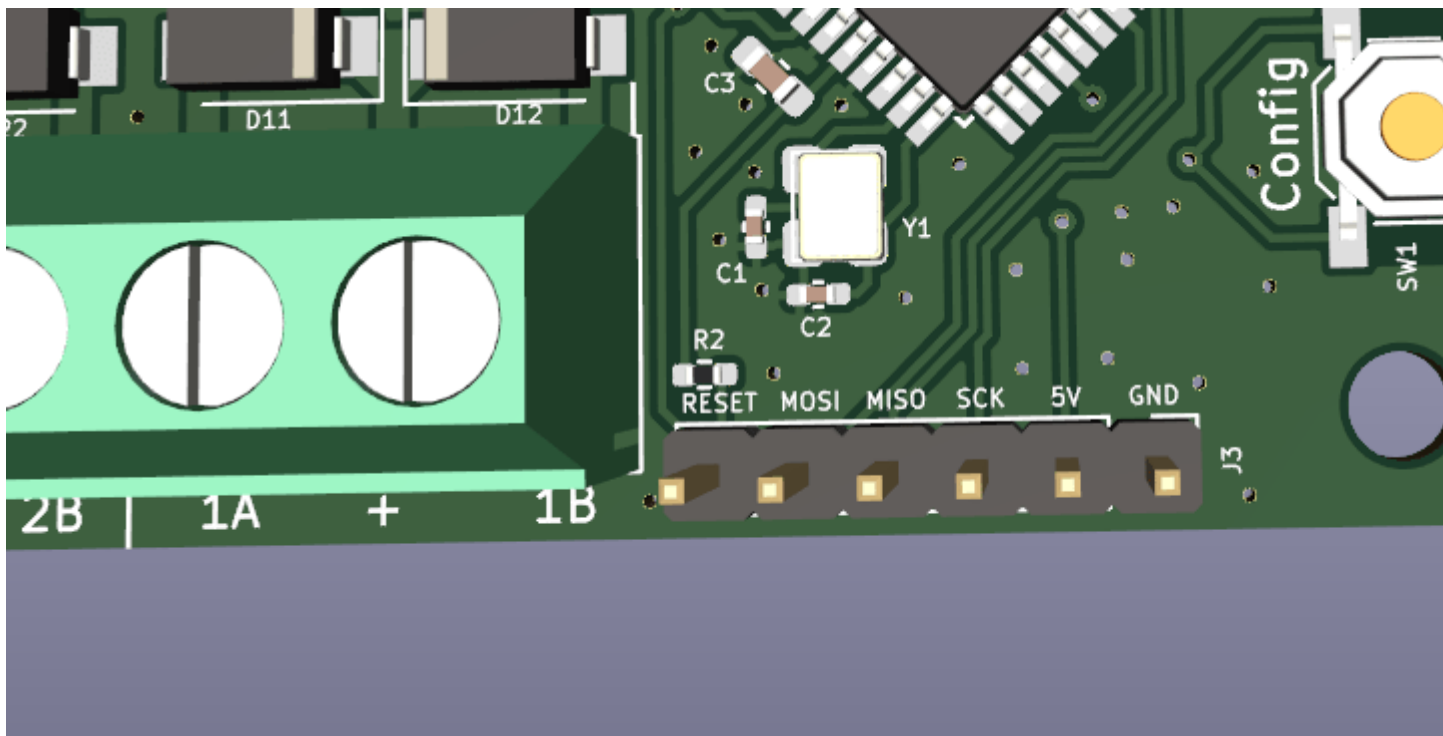
The first thing we must do is, is to make the connections between Arduino and decoder. Make sure that the Arduino is not plugged in the computer while you connect the wires.

Every decoder has an ICSP connector for programming. In order to connect the wires. You could solder a conventional pin connector on it. The pins are marked so it should be easy to hook it up to an arduino. You can simply use depont wires for this. Alternatively you can purchase a so called pogopin clamp. This allow you to program without the need of a soldering iron.

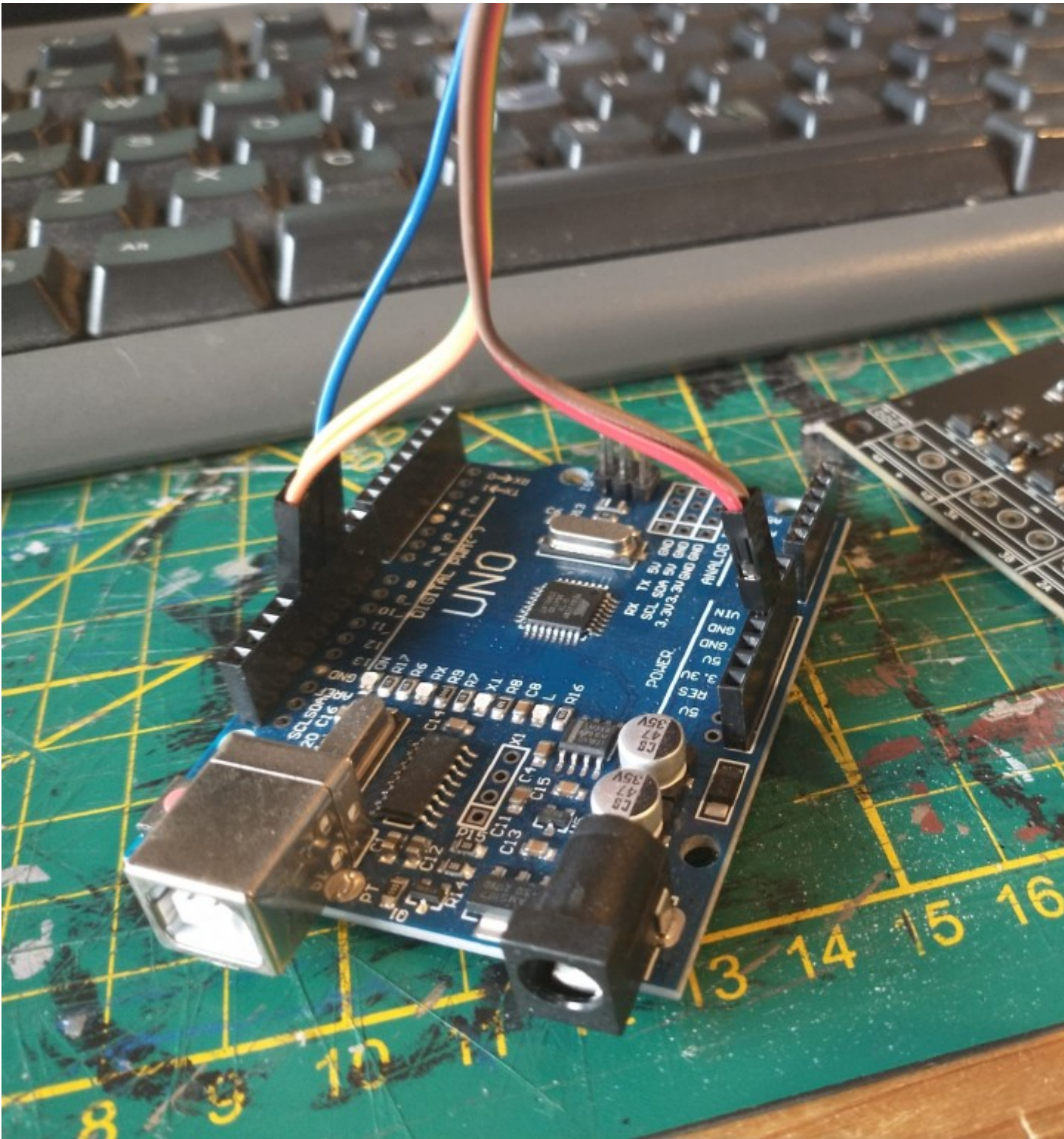
You need to make the following 6 connections between decoder (OSSD in this case) and Arduino board.

OSSD	UNO
5V	5V
GND	GND
SCK	D13
MISO	D12
MOSI	D11
RESET	D10

connections to make



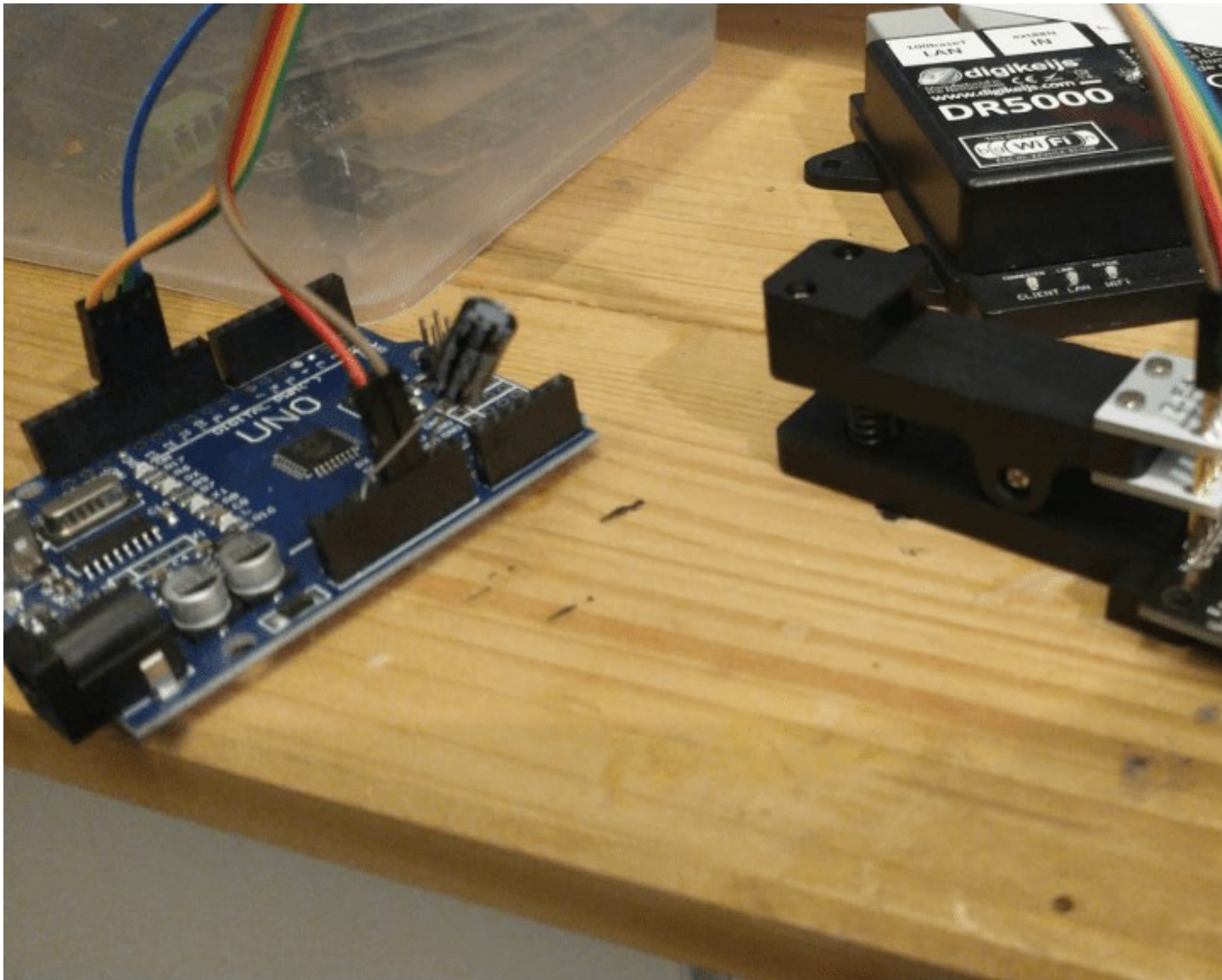
ICSP connector of the OSSD



An early OSSD model connected to the Arduino programmer.

When the Arduino is connected to the decoder you can plug it in the computer via an USB cable.

If you intend to do this often you may consider buying a 1×6 pogo pin clamp in 0.1" or 2.54mm pitch. With it you don't have to 'do' the wires and solder the program connector. This clamp has conical pins which are pressed against the program connector with a spring.

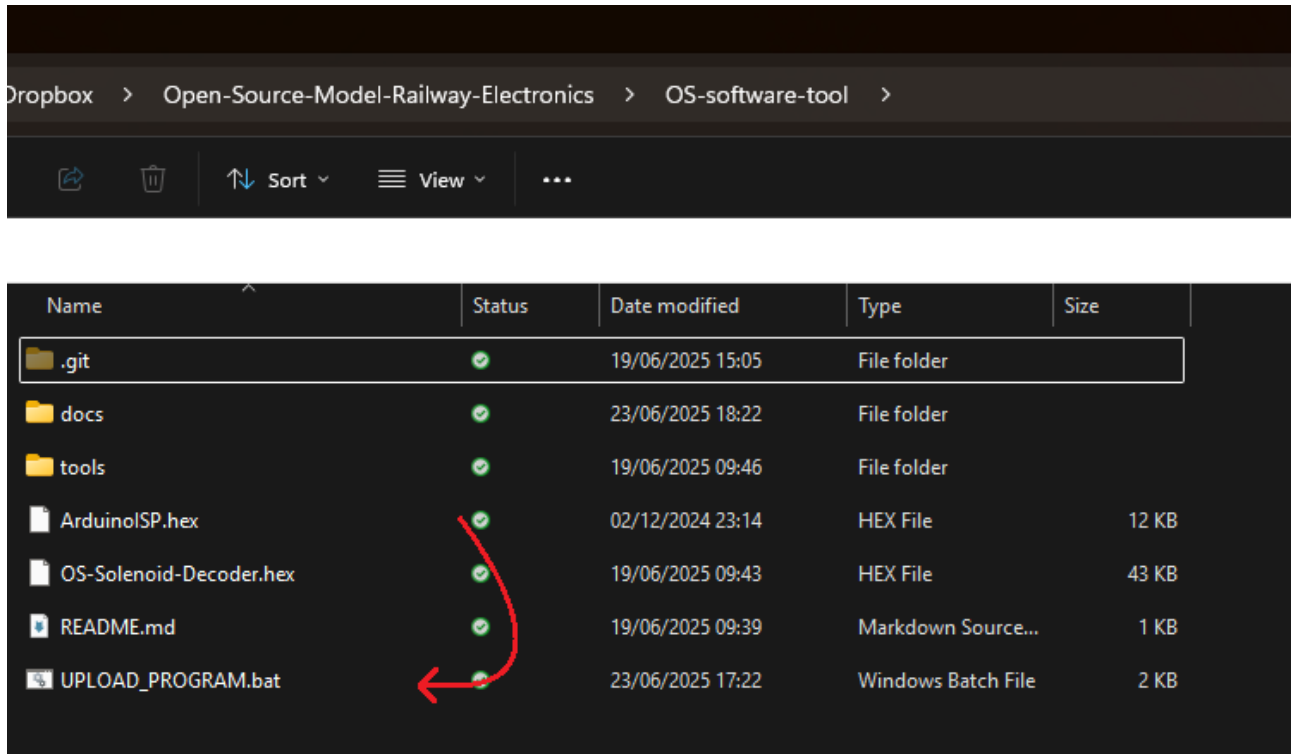


An fully assembled OSSD prorammed via pogopin clamp.

Flash the ArduinoISP program

The second thing we need to do is turn this Arduino board into a programmer by flashing Arduino's 'ArduinoISP' program into the board. When that is done we can use the Arduino board to program the OSSD.

For both programming the Arduino and a decoder there is an easy way and a slightly less easy way. The easy way is as simple as dragging "ArduinoISP.hex" over UPLOAD_PROGRAM.bat



The upload_program batch file can automatically detect an arduino when it is connected to to the PC via USB. It will then flash the hex file into the arduino, turning it into a fully pledged programmer device.

This was the easy way, drag one file over another. It could be however that something goes wrong and it does not work for whatever reason. In this case we can fall back to plan B

The plan B way involves downloading and installing the Arduino IDE on your computer. It is a fairly simple process you can just surf to arduino.cc, click “software” and navigate down to Arduino IDE 1.8.19

Legacy IDE (1.8.X)



Arduino IDE 1.8.19

The open-source Arduino Software (IDE) makes it easy to write code and upload it to the board. This software can be used with any Arduino board.

Refer to the [Getting Started](#) page for Installation instructions.

SOURCE CODE

Active development of the Arduino software is [hosted by GitHub](#). See the instructions for [building the code](#). Latest release source code archives are available [here](#). The archives are PGP-signed so they can be verified using [this](#) gpg key.

DOWNLOAD OPTIONS

Windows Win 7 and newer 

Windows ZIP file

Windows app Win 8.1 or 10 [Get](#) 

Linux 32 bits

Linux 64 bits

Linux ARM 32 bits

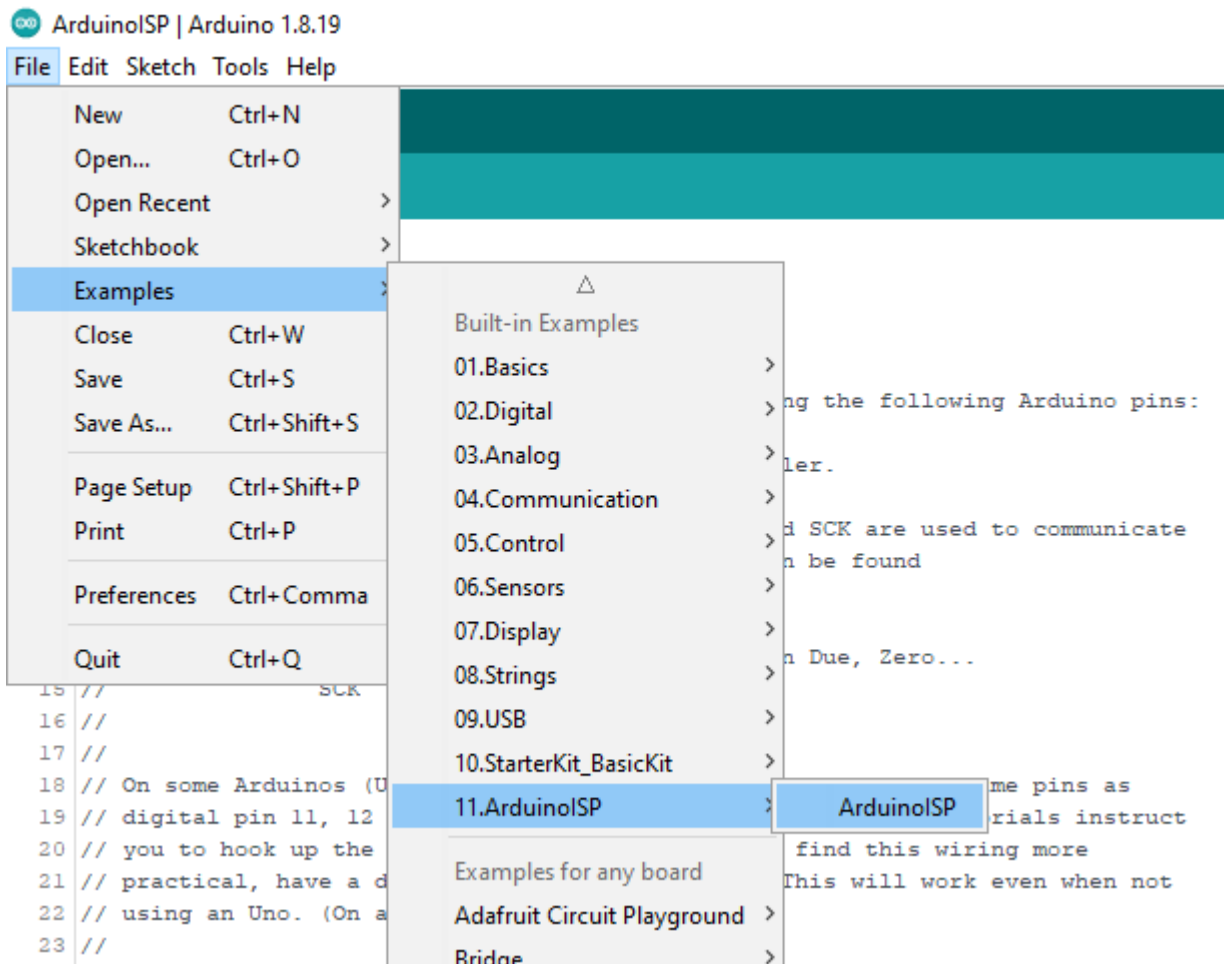
Linux ARM 64 bits

Mac OS X 10.10 or newer

[Release Notes](#)

[Checksums \(sha512\)](#)

Once installed you need to open the example sketch named ArduinoISP and flash it into the Arduino board.

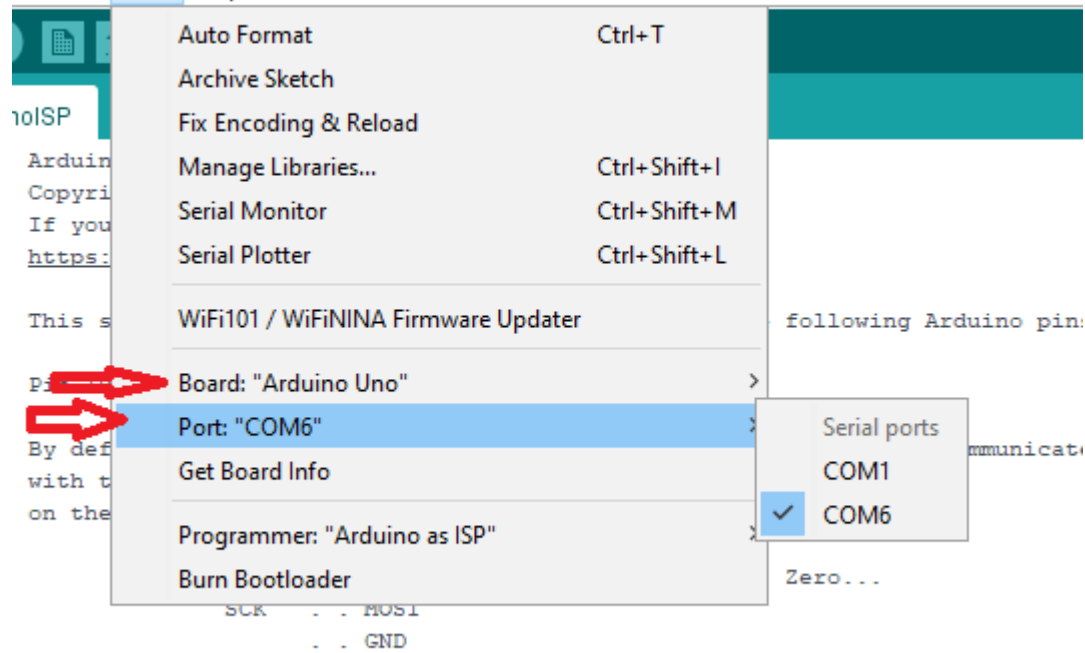


Open ArduinoISP sketch in the IDE

Make sure to set the COM port and board settings right. You can see which COM ports are available in the IDE. It is usually the bottom one. Set board type to UNO (or NANO depending on what you bought)

inoISP | Arduino 1.8.19

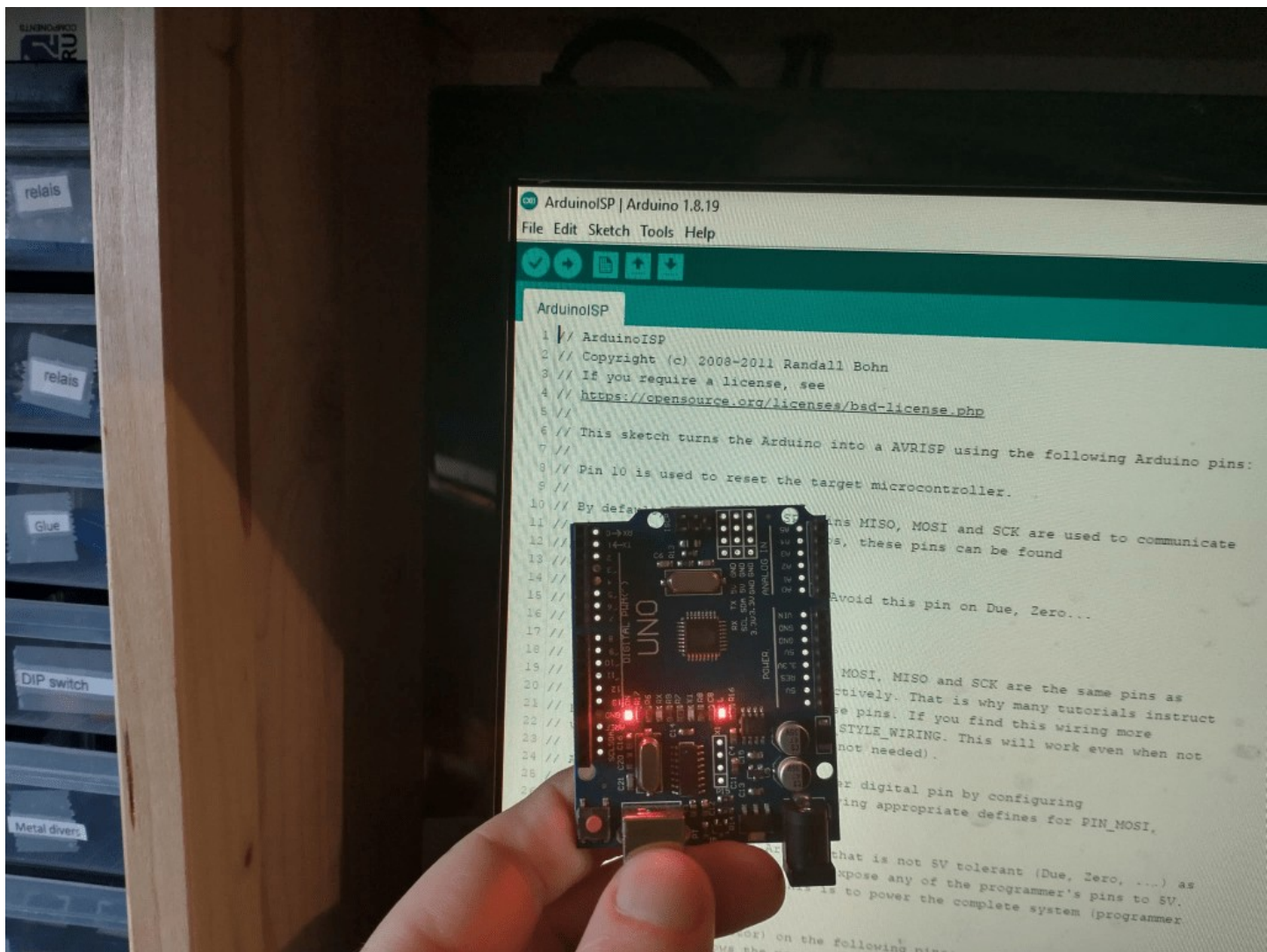
: Sketch Tools Help



On some Arduinos (Uno,...), pins MOSI, MISO and SCK are the same pins as

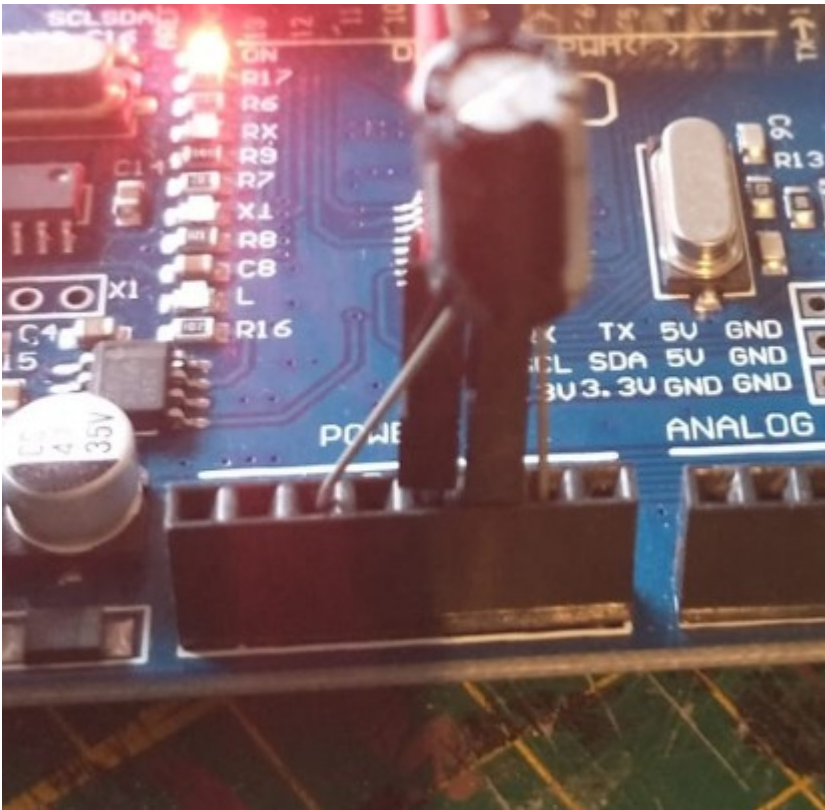
Set Board and COM port settings right.

Finally click on the upload arrow or hit Ctrl + U



Arduino UNO being programmed as ICSP programmer

When the Arduino is programmed it is almost ready to be used as programmer. You need to put in a capacitor in the Arduino (10uF or more). The long leg (+) should go in the reset pin and the short pin (-) should go in a GND pin. This capacitor prevents the Arduino from being reset when we try to program the decoder.

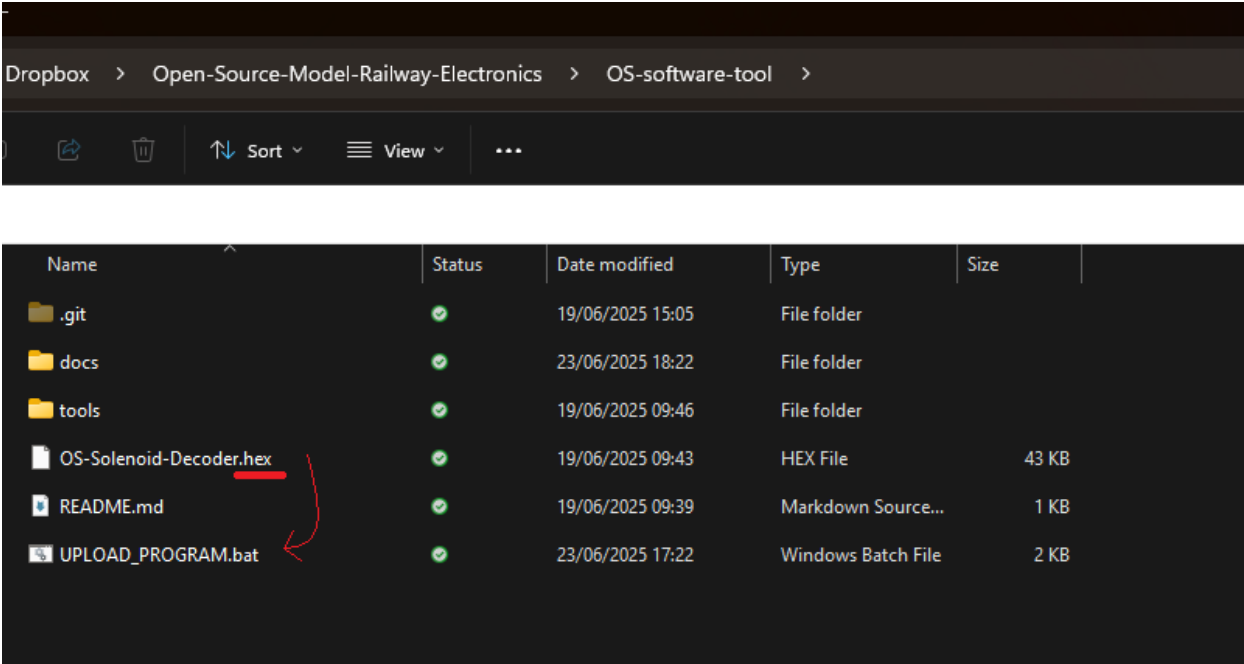


Added capacitor

Once you have successfully programmed the Arduino board with the ISP program, made all the proper connections and inserted the capacitor, the programming of the decoder can now commence!

Flash the Decoder

Like before there is the easy way and the plan B way. Again the easy way exists out of dragging a hex file over the upload_program.bat script. If that goes well you should see a similar result as before.



Name	Status	Date modified	Type	Size
.git	✓	19/06/2025 15:05	File folder	
docs	✓	23/06/2025 18:22	File folder	
tools	✓	19/06/2025 09:46	File folder	
OS-Solenoid-Decoder.hex	✓	19/06/2025 09:43	HEX File	43 KB
README.md	✓	19/06/2025 09:39	Markdown Source...	1 KB
UPLOAD_PROGRAM.bat	✓	23/06/2025 17:22	Windows Batch File	2 KB

```
C:\WINDOWS\system32\cmd.exe

C:\Users\Gebruiker\Documents\PROJECTS\Train-Science-DIY\OSSD\Software>echo off
select comport number
6
UPLOADING OSSD.hex on COM6

avrdude: AVR device initialized and ready to accept instructions

Reading | ##### | 100% 0.03s

avrdude: Device signature = 0x1e950f (probably m328p)
avrdude: reading input file "bin\OSSD.hex"
avrdude: writing flash (16196 bytes):

Writing | ##### | 100% 18.71s

avrdude: 16196 bytes of flash written
avrdude: reading input file "0xFD"
avrdude: writing efuse (1 bytes):

Writing | ##### | 100% 0.01s

avrdude: 1 bytes of efuse written
avrdude: reading input file "0xDE"
avrdude: writing hfuse (1 bytes):

Writing | ##### | 100% 0.01s

avrdude: 1 bytes of hfuse written
avrdude: reading input file "0xFF"
avrdude: writing lfuse (1 bytes):

Writing | ##### | 100% 0.01s

avrdude: 1 bytes of lfuse written

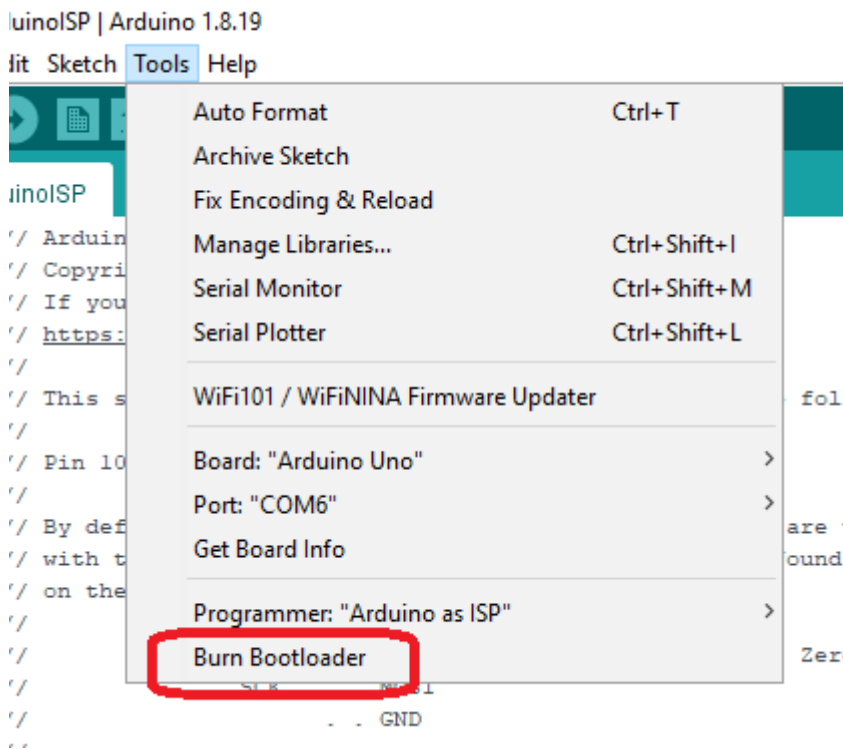
avrdude: safemode: Fuses OK (E:FD, H:DE, L:FF)

avrdude done. Thank you.

Press any key to continue . . .
```

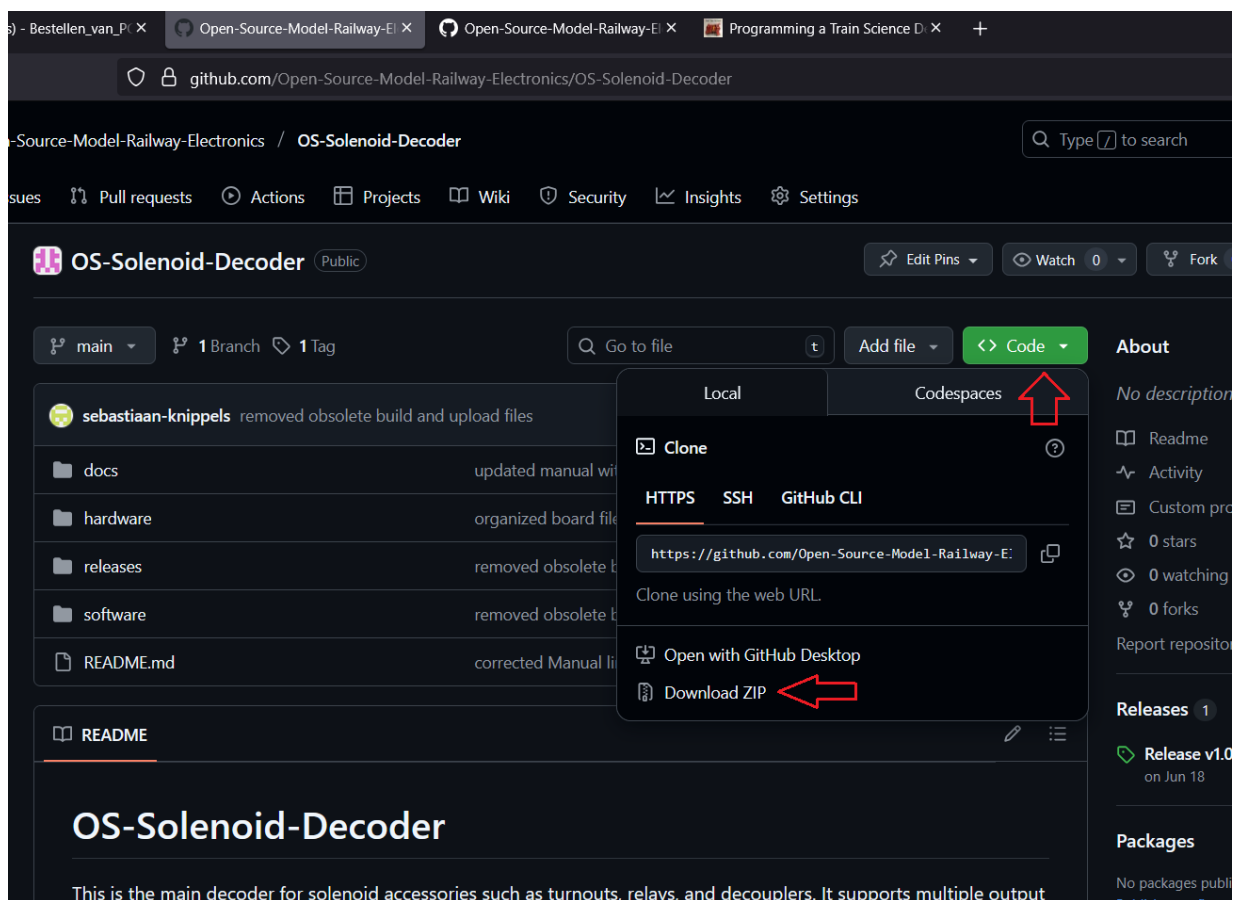
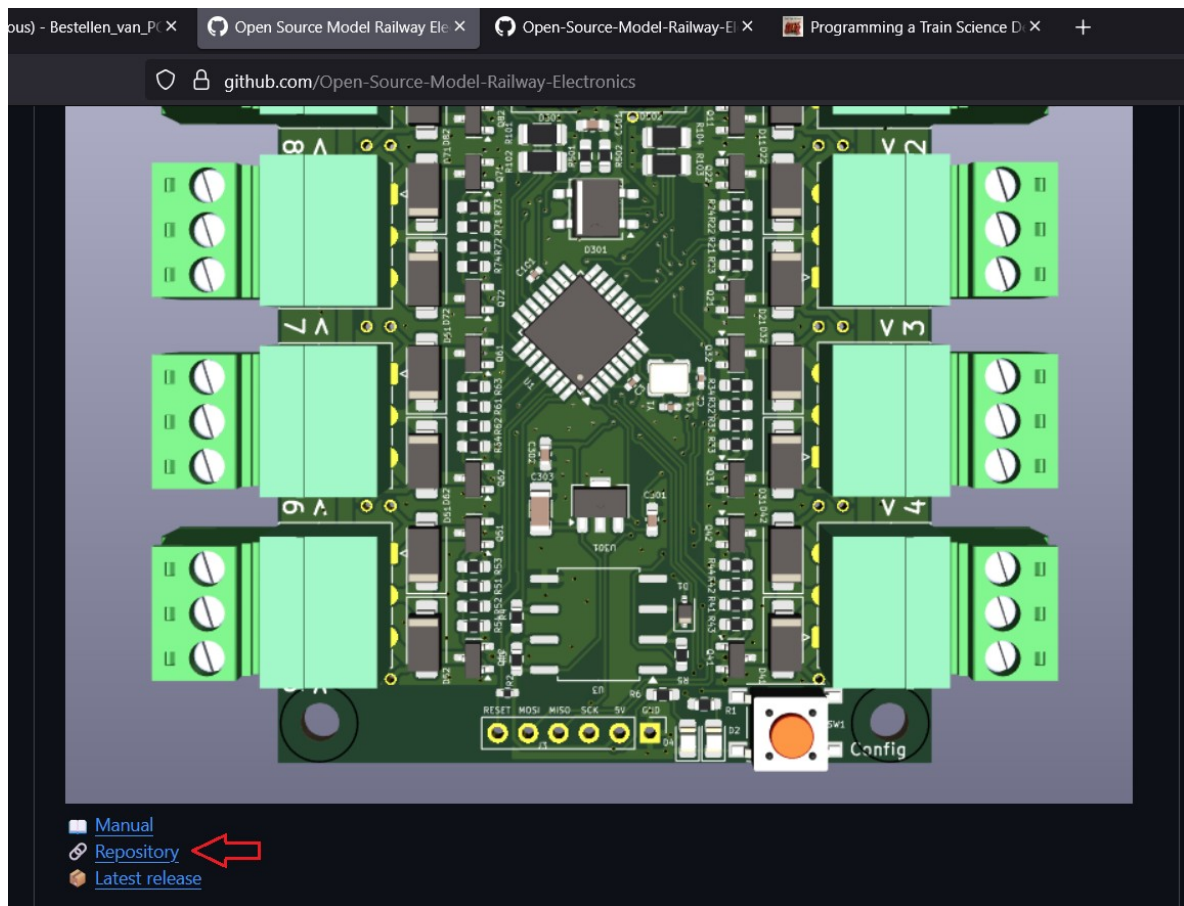
Note #2. It can occasionally happen that the upload process sometimes fails. Sometimes it tosses an error sometimes it does not an error and the board won't work. When it happens, just keep trying it again until it does work

With the Arduino IDE method (plan B), you need to do 2 steps. Before you program the board with the a Decoder program, you'll first need to burn the bootloader. This is also explained on the [Arduino website](#). It is however incredibly simple... You can just click on 'Tools' and 'burn bootloader'. Also make sure that board and COM port settings are correct.

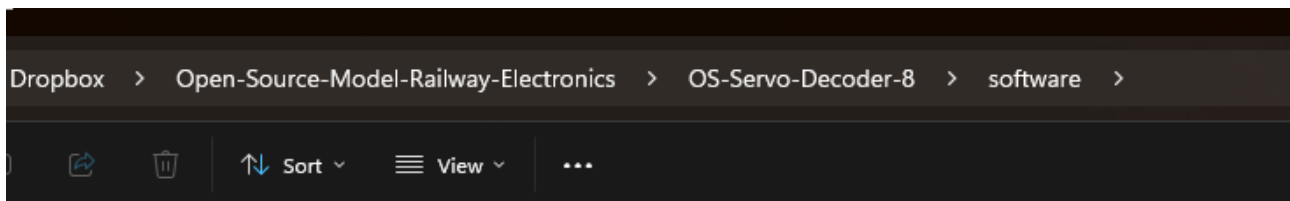


Extra information: The bootloader is usually needed for an arduino to allow programming via USB. However it is also important because it sets the Atmega's fuse bits to work with the correct clock frequency. If you don't do this the decoder will operate at 1/16 speed and won't be able to do decode a single DCC message.

When the bootloader is 'burned' you must open a decoder project in the Arduino IDE. For this to happen, you do not need the binary hex file, but the actual source code. On github navigate to the appropriate decoder, go to it's repository, press the green 'code' button and click download zip.



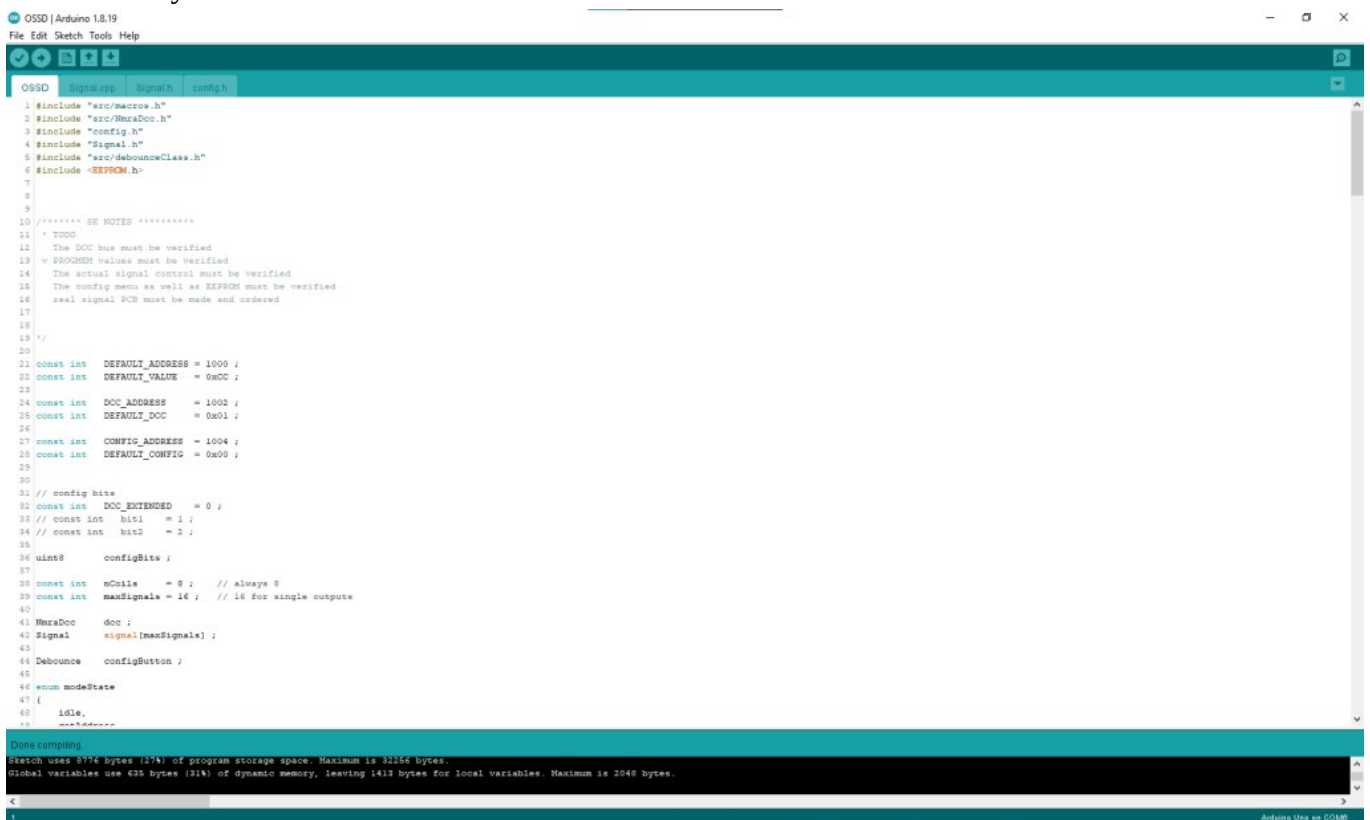
Afterwards unzip the folder. You should then see a software folder. In it there is a file called software.ino



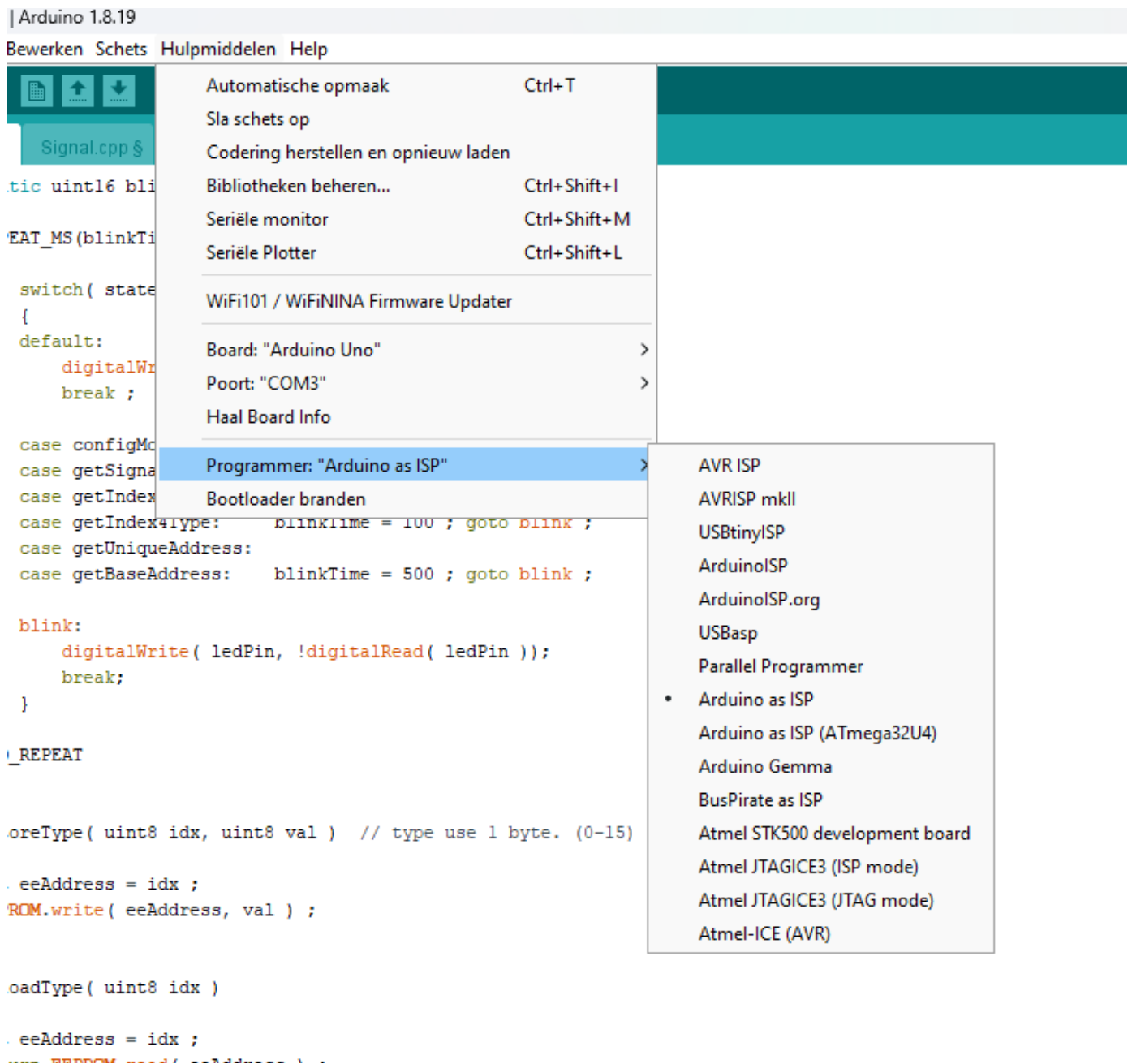
Name	Status	Date modified	Type	Size
.vscode	✓	27/10/2025 09:09	File folder	
build	✓	25/06/2025 16:38	File folder	
src	✓	27/10/2025 09:09	File folder	
config.h	✓	22/07/2025 16:23	C Header Source F...	1 KB
README.md	✓	18/06/2025 09:59	Markdown Source...	1 KB
software.ino	✓	23/07/2025 10:32	Arduino file	24 KB

Double click on the software.ino file. This should open the IDE for you with the desired program open.

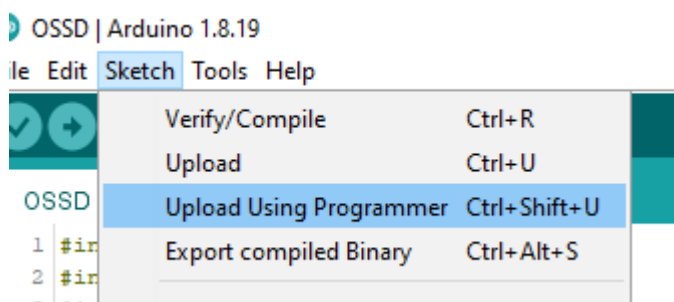
This is what you should see now.



In order to program a Decoder program into the Decoder we won't be using the conventional upload method with the arrow. You need to tell the IDE that we will be using an Arduino as ISP programmer. You can do so by clicking on 'Tools' -> 'Programmer' -> 'Arduino as ISP'.



Then you must click on 'Sketch' and 'Upload Using Programmer'. Or simply hit the shortcut Ctrl + Shift + U.



Doing so will let the Arduino IDE first compile the program and flash it into the Decoder. If you see this message in the bottom of the IDE.

Done uploading.

Sketch uses 8776 bytes (27%) of program storage space. Maximum is 32256 bytes.

Global variables use 635 bytes (31%) of dynamic memory, leaving 1413 bytes for local variables. Maximum